ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 2

Apply the Clustering Techniques Using Unsupervised Learning

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To apply an unsupervised learning clustering technique to group observations from the Diabetes dataset into similar clusters using the K-Means clustering algorithm.

2. Objectives

- To understand the concept of unsupervised learning and clustering.
- To preprocess and standardize the Diabetes dataset for clustering.
- To apply the K-Means clustering algorithm using different values of K.
- To evaluate the clustering results using inertia and silhouette score.
- To visualize the obtained clusters using PCA.
- To analyze the characteristics of the resulting clusters.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations and 10 independent variables. Since clustering is an unsupervised learning task, the target variable is excluded from the clustering process. The ten independent variables are standardized before applying K-Means so that features are considered on a comparable scale.

Property	Value
Number of samples	442
Clustering features	10
Target used for clustering	Not used
Algorithm	K-Means

4. Theory

Unsupervised learning is a machine learning approach in which the model learns patterns from data without using predefined class labels. Clustering is one of the main techniques of unsupervised learning and aims to divide observations into groups whose members are more similar to each other than to members of other groups.

K-Means clustering is a partition-based clustering algorithm. It selects K cluster centroids and repeatedly assigns each observation to the nearest centroid. The centroids are then updated as the mean of the observations assigned to each cluster. The process continues until the cluster assignments or centroid positions stabilize.

The main objective of K-Means is to minimize within-cluster variation. This is commonly represented by inertia, which is the sum of squared distances between observations and their assigned cluster centers. Lower inertia indicates more compact clusters, although inertia normally decreases as K increases.

The silhouette score measures how well an observation fits within its assigned cluster compared with other clusters. It ranges from -1 to +1. Higher values generally indicate better separated and more cohesive clusters.

Cluster profiling is another useful step after clustering. The average standardized feature values of each cluster can be compared to understand how the groups differ. A cluster profile heatmap provides a compact visual representation of these differences and helps interpret the characteristics of each group.

PCA (Principal Component Analysis) is used in this experiment only for visualization. Since the dataset contains ten dimensions, it is difficult to visualize the clusters directly. PCA projects the standardized data onto a smaller number of principal components while retaining as much variance as possible. The first two principal components can then be plotted to obtain a two-dimensional view of the cluster structure.

In this experiment, the target variable is not used during clustering because the objective is to perform unsupervised learning. The ten independent variables are standardized and supplied to K-Means. The target is therefore kept separate from the clustering process.

Feature scaling is important before applying K-Means because the algorithm is distance-based. If one feature has a much larger numerical scale than another, it can dominate the distance calculation. Standardization transforms features so that they have approximately zero mean and unit standard deviation, allowing the features to contribute more comparably to the clustering process.

The silhouette score ranges from -1 to +1. A value close to +1 indicates that observations are well matched to their own clusters and well separated from other clusters. A value close to 0 indicates overlapping clusters, while a negative value may indicate that some observations have been assigned to an inappropriate cluster.

However, inertia alone cannot be used to select the ideal number of clusters because it generally decreases when K increases. Therefore, clustering quality can also be evaluated using the silhouette score. The silhouette score compares the average distance of an observation from points in its own cluster with its distance from points in the nearest neighboring cluster.

The main objective of K-Means is to minimize the within-cluster sum of squared errors. This objective is also represented by the inertia of the clustering solution. Inertia is calculated as the sum of squared distances between every observation and the centroid of the cluster to which it is assigned. Lower inertia indicates more compact clusters.

For two observations X and Y, Euclidean distance can be expressed as: $d(X,Y) = \sqrt{\Sigma(X_i - Y_i)^2}$. A smaller distance indicates greater similarity between observations. In K-Means, this distance is used to determine the nearest cluster centroid.

The K-Means algorithm starts by selecting K initial centroids. Each observation is then assigned to the nearest centroid, usually using Euclidean distance. After the assignments are made, the centroid of every cluster is recalculated as the mean of all observations assigned to that cluster. The assignment and centroid-update steps are repeated until the centroids or cluster assignments become stable.

K-Means is one of the most widely used partitioning-based clustering algorithms. The value of K represents the number of clusters that the algorithm is expected to create. For each cluster, K-Means maintains a centroid, which represents the mean position of the observations belonging to that cluster.

Clustering is an unsupervised machine learning technique used to discover natural groups or patterns in a dataset. Unlike supervised learning, clustering does not require a predefined target or class label during model training. The algorithm instead uses similarities or distances between observations to form groups called clusters.

5. Methodology

- Load the Diabetes dataset using scikit-learn.
- Remove the continuous target variable because clustering is performed without using class labels.
- Standardize the ten independent variables using StandardScaler.
- Apply K-Means clustering for K values from 2 to 6.
- Calculate inertia and silhouette score for each K.
- Select the K value giving the highest silhouette score among the tested values.
- Use PCA only to project the clustered data into two dimensions for visualization.
- Analyze the cluster profiles using the mean standardized feature values.

6. Outputs

The following outputs were generated from the clustering experiment.

6.1 Cluster Evaluation

K	Inertia	Silhouette Score
2	3176.6549	0.2370
3	2879.2112	0.1535
4	2642.6894	0.1372
5	2453.2386	0.1467
6	2307.6072	0.1430

The highest silhouette score among the tested values was obtained for K = 2, with a silhouette score of 0.2370. The corresponding inertia was 3176.6549.

6.2 Cluster Evaluation Graph

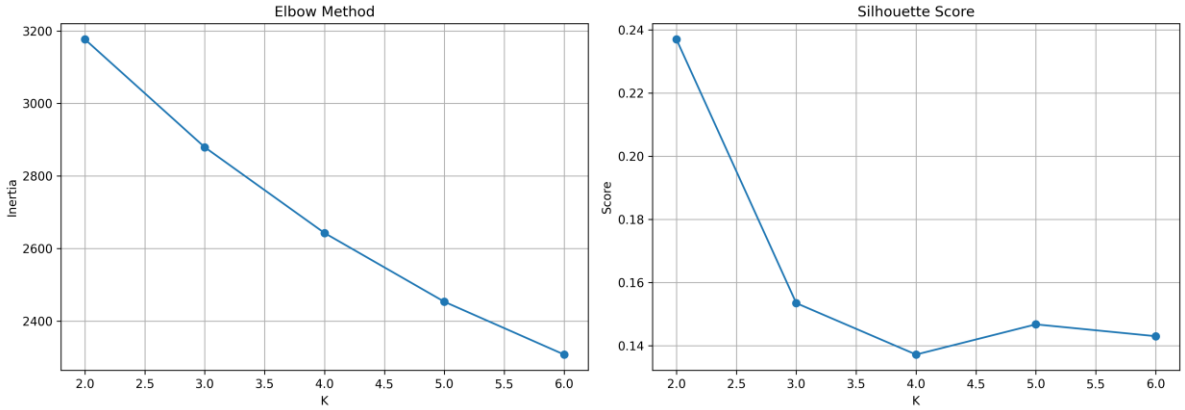


Figure 1: Inertia and silhouette score for different K values

6.3 PCA Cluster Visualization

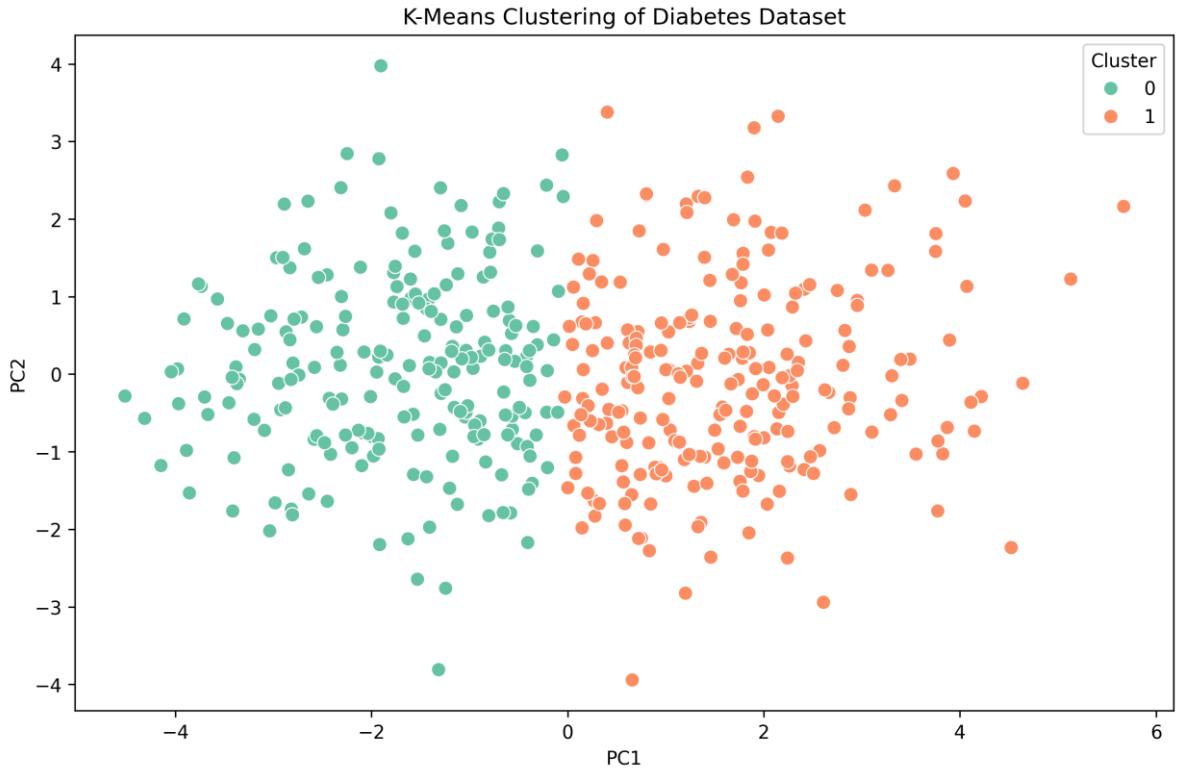


Figure 2: PCA visualization of K-Means clusters

6.4 Cluster Profiles

Cluster	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6
0.0	-0.0179	-0.0161	-0.0252	-0.0219	-0.0258	-0.0276	0.0264	-0.0355	-0.0319	-0.0261
1.0	0.0168	0.0151	0.0236	0.0206	0.0242	0.0259	-0.0248	0.0334	0.0300	0.0245

6.5 Cluster Profile Heatmap

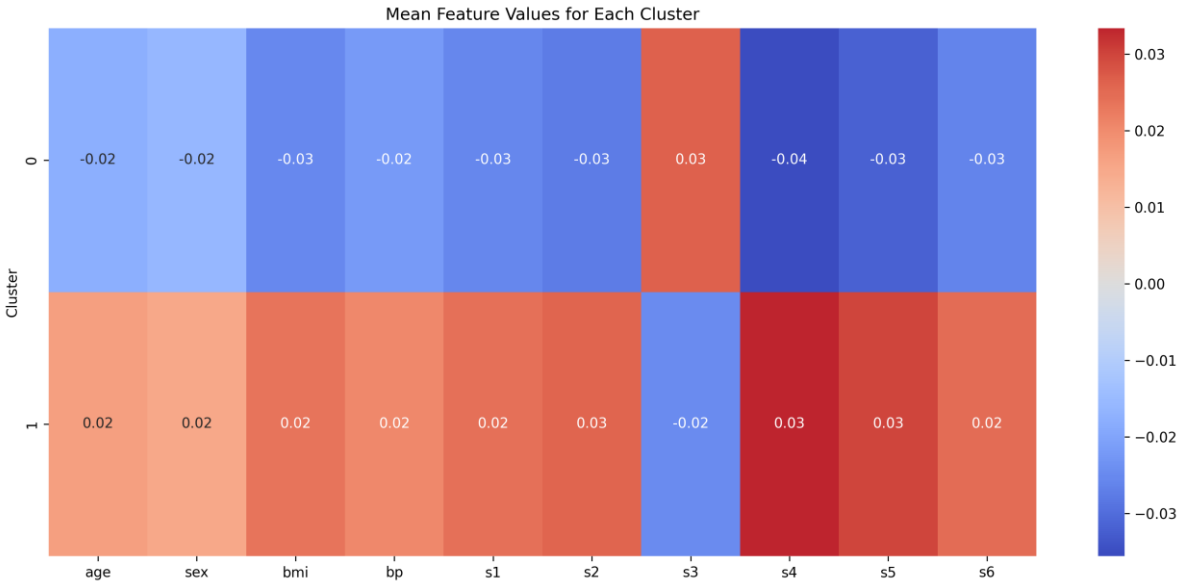


Figure 3: Heatmap of standardized cluster profiles

7. Result

K-Means clustering was successfully applied to the ten independent variables of the Diabetes dataset. Clustering was evaluated for K values from 2 to 6. Based on the highest silhouette score, K = 2 was selected as the best configuration among the tested values. The selected solution produced a silhouette score of 0.2370.

8. Observations

- K = 2 produced the highest silhouette score (0.2370) among the tested K values.
- Inertia decreases as the number of clusters increases because additional clusters reduce within-cluster distances.
- The silhouette score is useful for comparing cluster separation and cohesion.
- PCA was used only for two-dimensional visualization; the clustering itself used all ten standardized features.
- The cluster profile heatmap shows the relative standardized feature values of the obtained clusters.
- The two clusters show opposite average patterns across many standardized features, while s3 follows the opposite direction.

9. Conclusion

The experiment successfully demonstrated unsupervised learning using K-Means clustering. The Diabetes dataset was standardized and clustered without using the target variable. Among K values from 2 to 6, K = 2 gave the best silhouette score. The evaluation metrics and visualizations helped in understanding cluster quality, while PCA provided a two-dimensional representation of the resulting clusters.